Finance Stochast. 6, 197–225 (2002)

Finance and
Stochastics

On Lévy processes, Malliavin calculus and market models with jumps

Jorge A. León[1], **Josep L. Solé**[2], **Frederic Utzet**[2], **Josep Vives**[2]

[1] Departamento de Matemáticas, CINVESTAV-IPN, Apartado Postal 14-740, 07000 México, D.F., México (e-mail: jleon@math.cinvestav.mx)
[2] Departament de Matemàtiques, Edifici Cc, Universitat Autònoma de Barcelona, 08193 Bellaterra (Barcelona) Spain (e-mail: jllsole@mat.uab.es, utzet@mat.uab.es, vives@mat.uab.es)

Abstract. Recent work by Nualart and Schoutens (2000), where a kind of chaotic property for Lévy processes has been proved, has enabled us to develop a Malliavin calculus for Lévy processes. For simple Lévy processes some useful formulas for computing Malliavin derivatives are deduced. Applications for option hedging in a jump–diffusion model are given.

Key words: Lévy processes, Malliavin calculus, jump diffusion model, option hedging

JEL classification: G12

Mathematics Subject Classification (1991): 60H40, 60G20

Introduction

This paper has two objectives. The first one is to develop the initial steps of Malliavin Calculus for Lévy Processes. The Malliavin calculus in the Brownian setup has two approaches which turn out to be equivalent: one as a weak derivative in canonical space and the other one through Wiener chaos (see Nualart 1995 for a complete account of this theory). In general, a Lévy process has no chaotic decomposition property in the sense that Brownian motion, Poisson process, or so-called normal martingales have (see Ma et al. 1998), but recent work (Nualart and Schoutens 2000), where a kind of chaotic representation property for Lévy processes has been proved, has enabled us to define a Malliavin derivative using the chaotic approach. However, for present purposes, the Lévy processes studied by Nualart and Schoutens (2000) are too general and we will restrict ourselves to

Correspondence to: J.L. Solé

Manuscript received: November 2000; final version received: May 2001

a brief investigation of their most relevant properties. We center our research on a very simple Lévy process –the sum of a Brownian motion and k independent Poisson processes– for which is possible to obtain a weak derivative interpretation and useful formulas. More general Lévy processes, such as the sum of a Brownian motion and a compound Poisson process can be approximated by these simple Lévy processes. In preparing this paper we came across a paper by Løkka (1999) which containes several extremely useful ideas.

The second goal of this paper is option hedging in a jump-diffusion model. Models with jumps for a market are an old topic in Mathematical Finance; The initial paper of Black–Scholes (1973) was rapidily followed by Merton (1976) where the first jump–diffusion model was proposed. Here we approximate a jump–diffusion model for a simple Lévy process of the type studied in the first part of the paper, and using a Malliavin Calculus approach (see Øksendal 1996 for this technique in the Black-Scholes case) we hedge a european call. In this connexion we would like to acknowledge the paper by Jensen (1999) dealing with the problem of pricing a european call in a jump–diffusion model. A different approach for extending Clark-Haussman-Ocone formula using white noise analysis and its applications to mathematical finance can be found in Aase et al. (2000).

The present paper is organized as follows. The first section deals with the theory of Malliavin derivatives for a general Lévy process and we answer some questions that naturally arise from the paper by Nualart and Schoutens (2000). In the second section we center the results of Sect. 1 on more tractable Lévy processes that we call *simple Lévy processes* and we obtain some interesting formulas to compute Malliavin derivatives. In Sect. 3, we prove some results in order to approximate the Lévy process used as a jump–diffusion model by means of simple Lévy processes. Finally, in Sect. 4, we obtain formulas which enables an approximate hedge of a european call in a jump–diffusion model.

1 General theory

1.1 Notations

Let $X = \{X_t,\ t \geq 0\}$ be a Lévy process (that is, X has stationary and independent increments, is continuous in probability and $X_0 = 0$), and henceforth we always assume that we are using the *cadlag* version) on a complete probability space $(\Omega, \mathscr{F}, P)$, and let $\{\mathscr{F}_t,\ t \geq 0\}$ be the natural filtration of X completed with the null sets of $\mathscr{F}$. We also assume that the Lévy measure ν of X satisfies that there exist $\varepsilon > 0$ and $\delta > 0$ such that

$$\int_{(-\varepsilon,\varepsilon)^c} e^{\delta\,|x|}\,\nu(dx) < \infty. \tag{1.1}$$

This implies that X_t has moments of all orders and that the polynomials are dense in $L^2(\mathbb{R}, P \circ X_1^{-1})$ (see Nualart and Schoutens 2000)

Define

$$\begin{aligned} X_t^{(1)} &= X_t, \\ X_t^{(i)} &= \sum_{0<s\leq t} (\Delta X_s)^i, \quad i \geq 2. \end{aligned}$$

We have:

– The processes $X^{(i)} = \{X_t^{(i)}, t \geq 0\}$, $i = 1, 2, \ldots$, are Lévy processes that jump at the same points as X.
– $E(X_t^{(i)}) = m_i t$, where $m_1 = E(X_1)$ and $m_i = \int_{-\infty}^{\infty} x^i \, \nu(dx)$, $i \geq 2$.

Now define the processes

$$Y_t^{(i)} = X_t^{(i)} - m_i \, t, \quad i \geq 1.$$

– The processes $Y^{(i)} = \{Y_t^{(i)}, t \geq 0\}$ are martingales. The predictable quadratic covariation process of $Y^{(i)}$ and $Y^{(j)}$ is given by

$$\langle Y^{(i)}, Y^{(j)} \rangle_t = m_{i+j} \, t, \quad i, j \geq 2.$$

– The quadratic covariation of $Y^{(i)}$ and $Y^{(j)}$ is given by

$$[Y^{(i)}, Y^{(j)}]_t = X_t^{(i+j)} + \mathbf{1}_{\{i=j=1\}} \sigma^2 t, \quad i, j \geq 1,$$

where σ is a parameter corresponding to the gaussian part of the process X (see Nualart and Schoutens (2000) for the notation.)

Now, we introduce the processes

$$H_t^{(i)} = \sum_{j=1}^{i} a_{ij} Y_t^{(j)}, \quad i \geq 1, \tag{1.2}$$

where the constants a_{ij} are chosen in such a way that $a_{i1} = 1$ and the martingales $H^{(i)}$, $i = 1, 2, \ldots$ are pairwise strongly orthogonal, that means, for $i \neq j$, the process $H^{(i)} H^{(j)}$ is a martingale. In particular, since $\langle H^{(i)}, H^{(j)} \rangle_t$ is a predictable process such that $H_t^{(i)} H_t^{(j)} - \langle H^{(i)}, H^{(j)} \rangle_t$ is a martingale, we have that

$$\langle H^{(i)}, H^{(j)} \rangle_t = 0, \quad i \neq j.$$

Moreover,

– The processes $\{H_t^{(i)}, t \geq 0\}$ are martingales with predictable quadratic variation process given by

$$\langle H^{(i)}, H^{(i)} \rangle_t = q_i \, t,$$

where

$$q_i = \sum_{j, j'=1, \ldots, i} a_{ij} a_{ij'} m_{j+j'} + a_{i1}^2 \sigma^2.$$

– The quadratic covariation of $H^{(i)}$ and $H^{(j)}$ is given by

$$[H^{(i)}, H^{(j)}]_t = \sum_{k=1}^{i} \sum_{k'=1}^{j} a_{ik} a_{jk'} X_t^{(k+k')} + a_{i1} a_{j1} \sigma^2 t.$$

1.2 Iterated integrals

In order to define a stochastic integral with respect to either Gaussian processes, Poisson processes or normal martingales (Ma et al. 1998), it is enough to deal with multiple integrals. However, for Lévy processes, we use iterated integrals (instead of multiple ones) because of the chaotic representation given by Nualart and Schautens (2000), which involves the family $H^{(i)},\ i = 1, 2, \ldots$ as integrators. In this case, the basic property of Itô multiple integrals

$$I_n(f) = I_n(\widetilde{f})$$

($\widetilde{f}$ is the symmetrized of the function f) is lost if we are integrating with respect to different $H^{(1)}, \ldots, H^{(n)}$ instead of only one integrator.

Let $\Sigma_n = \{(t_1, \ldots, t_n) \in \mathbb{R}_+^n : 0 < t_1 < t_2 < \cdots < t_n\}$ be the positive simplex of $\mathbb{R}^n$. Given $f \in L^2(\mathbb{R}_+^n)$ we will denote by $J_n^{(i_1,\ldots,i_n)}(f)$ the iterated integral of f with respect to $H^{(i_1)}, \ldots, H^{(i_n)}$:

$$\begin{aligned} J_n^{(i_1,\ldots,i_n)}(f) \quad = \quad & \int_0^{\infty} \left(\int_0^{t_n -} \cdots \left(\int_0^{t_2 -} f(t_1, \ldots, t_n) dH^{(i_1)}(t_1) \right) \cdots \right. \\ & \left. \cdot dH^{(i_{n-1})}(t_{n-1}) \right) dH^{(i_n)}(t_n). \end{aligned}$$

(Usually, we will omit all the parenthesis to simplify the notation). We remark that all these integrals are well defined since all the processes $H^{(i)},\ i = 1, 2, \ldots$ are square integrable martingales with respect to the filtration $\{\mathscr{F}_t,\ t \geq 0\}$.

In the remain of this paper, we use the notations of Ma et al. (1998) for the indices $(t_1, \ldots, t_n)$ in the simplex Σ_n, where $t_1 < \cdots < t_n$, instead of that of Nualart and Schoutens (2000) who write the indices in decreasing order.

Proposition 1.1 *Consider $f \in L^2(\mathbb{R}_+^n)$ and $g \in L^2(\mathbb{R}_+^m)$. Then*

$$\begin{aligned} & E\left[J_n^{(i_1,\ldots,i_n)}(f)\, J_m^{(j_1,\ldots,j_m)}(g)\right] \\ & = \begin{cases} q_{i_1} \cdots q_{i_n} \int_{\Sigma_n} & f(t_1, \ldots, t_n)\, g(t_1, \ldots, t_n)\, dt_1 \cdots dt_n, \\ & \qquad \text{if } n = m \text{ and } (i_1, \ldots, i_n) = (j_1, \ldots, j_n), \\ 0, & \text{otherwise}. \end{cases} \end{aligned}$$

Proof The definition of the iterated integral implies

$$\begin{aligned} & E\left[J_n^{(i_1,\ldots,i_n)}(f)\, J_m^{(j_1,\ldots,j_m)}(g)\right] = E\left[\langle J_n^{(i_1,\ldots,i_n)}(f),\ J_m^{(j_1,\ldots,j_n)}(g)\rangle_\infty\right] \\ & \quad = E\left[\int_0^\infty \left(\int_0^{t_n -} \cdots \int_0^{t_2 -} f(t_1, \ldots, t_{n-1}, t) dH^{(i_1)}(t_1) \cdots dH^{(i_{n-1})}(t_{n-1}) \right.\right. \\ & \qquad \left. \cdot \int_0^{t_m -} \cdots \int_0^{t_2 -} g(t_1, \ldots, t_{m-1}.t) dH^{(j_1)}(t_1) \cdots dH^{(j_{m-1})}(t_{m-1}) \right) \\ & \qquad \qquad \left. \cdot\, d\langle H^{(i_n)}, H^{(j_m)}\rangle(t) \right]. \end{aligned}$$

Hence the result follows using induction on n. □

The main results of the paper of Nualart and Schoutens (2000) are the chaotic and the predictable representation properties of the square integrable random variables:

Theorem 1.2 *(Nualart and Schoutens) Let $F \in L^2(\Omega, \mathscr{F}, P)$. Then F has a unique representation of the form*

$$F = E[F] + \sum_{n=1}^{\infty} \sum_{i_1,\dots,i_n \geq 1} J_n^{(i_1,\dots,i_n)}(f_{i_1,\dots,i_n}),$$

where $f_{i_i,\dots,i_j} \in L^2(\Sigma_j)$.

An immediate consequence of Theorem 1.2 is the predictable representation property of the square integrable random variables:

Theorem 1.3 *(Nualart and Schoutens) Let $F \in L^2(\Omega, \mathscr{F}, P)$. Then F has a representation of the form*

$$F = E[F] + \sum_{k=1}^{\infty} \int_0^{\infty} \phi_k(t)\, dH^{(k)}(t),$$

where $\{\phi_k(t),\, t \geq 0\},\ k \geq 1$, are predictable processes.

1.3 Derivative operators

This section is devoted to the study of some properties of the derivatives in the context of calculus of variations.

In the remaining of this paper, write

$$\Sigma_n^{(k)}(t) = \{(t_1, \dots, \widehat{t_k}, \dots, t_n) \in \Sigma_{n-1} : 0 < t_1 < \cdots < t_{k-1} < t \leq t_{k+1} < \cdots < t_n\}$$

and $\widehat{i}$ means that the i-th index is omitted. Observe that if $k \neq k'$ then $\Sigma_n^{(k)}(t) \cap \Sigma_n^{(k')}(t) = \emptyset$.

Definition 1.4 *Let $f \in L^2(\mathbb{R}_+^n)$ and $\ell \geq 1$. The process*

$$D_t^{(\ell)} J_n^{(i_1,\dots,i_n)}(f) = \sum_{k=1}^{n} \mathbf{1}_{\{i_k=\ell\}} J_{n-1}^{(i_1,\dots,\widehat{i_k},\dots,i_n)} \left(f(\underbrace{\cdots}_{k-1}, t, \cdots) \mathbf{1}_{\Sigma_n^{(k)}(t)}(\cdot) \right),$$

is called the derivative of $J_n^{(i_1,\dots,i_n)}(f)$ in the ℓ–th direction.

In order to clarify this definition, we will analyze explicitly the derivative of a iterated integral of order 3, and we will also check that our definition coincides with the Malliavin derivative in the terms of Ma et al. (1998) when $H^{(i_1)} = \cdots = H^{(i_n)}$, and also with the classical Malliavin derivative for the Brownian case (Nualart 1995).

Consider

$$J_3^{(i,j,k)}(f) = \int_0^\infty \left(\int_0^{w^-} \left(\int_0^{v^-} f(u,v,w) dH^{(i)}(u) \right) dH^{(j)}(v) \right) dH^{(k)}(w),$$

whose ℓ–th derivative is

$$\begin{aligned}
D_t^{(\ell)} J_3^{(i,j,k)}(f) &= \mathbf{1}_{\{k=\ell\}} J_2^{i,j} \left(f(\cdot,\cdot,t) \mathbf{1}_{\Sigma_3^{(3)}(t)}(\cdot,\cdot) \right) \\
&\quad + \mathbf{1}_{\{j=\ell\}} J_2^{i,k} \left(f(\cdot,t,\cdot) \mathbf{1}_{\Sigma_3^{(2)}(t)}(\cdot,\cdot) \right) \\
&\quad + \mathbf{1}_{\{i=\ell\}} J_2^{j,k} \left(f(t,\cdot,\cdot) \mathbf{1}_{\Sigma_3^{(1)}(t)}(\cdot,\cdot) \right) \\
&= \mathbf{1}_{\{k=\ell\}} \int_0^\infty \left(\int_0^{v^-} f(u,v,t) \mathbf{1}_{\{u<v<t\}} \, dH^{(i)}(u) \right) dH^{(j)}(v) \\
&\quad + \mathbf{1}_{\{j=\ell\}} \int_0^\infty \left(\int_0^{v^-} f(u,t,v) \mathbf{1}_{\{u<t\leq v\}} \, dH^{(i)}(u) \right) dH^{(k)}(v) \\
&\quad + \mathbf{1}_{\{i=\ell\}} \int_0^\infty \left(\int_0^{v^-} f(t,u,v) \mathbf{1}_{\{t\leq u<v\}} \, dH^{(j)}(u) \right) dH^{(k)}(v) \\
&= \mathbf{1}_{\{k=\ell\}} \int_0^{t^-} \left(\int_0^{v^-} f(u,v,t) \, dH^{(i)}(u) \right) dH^{(j)}(v) \\
&\quad + \mathbf{1}_{\{j=\ell\}} \int_t^\infty \left(\int_0^{t^-} f(u,t,v) \, dH^{(i)}(u) \right) dH^{(k)}(v) \\
&\quad + \mathbf{1}_{\{i=\ell\}} \int_t^\infty \left(\int_t^{v^-} f(t,u,v) \, dH^{(j)}(u) \right) dH^{(k)}(v).
\end{aligned}$$

Note that if $i = j = k = \ell$ and f is symmetric, then

$$D_t^{(i)} J_3^{(iii)}(f) = J_2^{(ii)} \left(f(\cdot,\cdot,t) \right).$$

In general, for a symmetric $f \in L^2(\mathbb{R}_+^n)$, we have

$$D_t^{(\ell)} J_n^{(\ell,\ldots,\ell)}(f) = J_{n-1}^{(\ell,\ldots,\ell)} \left(f_n(\cdot,t) \right).$$

In the Ma et al. (1998) setup, since $I_n(f) = n! J_n^{(\ell,\ldots,\ell)}(f)$, where $I_n(f)$ is the multiple integral of order n with respect to the martingale $H^{(\ell)}$, we obtain

$$D_t I_n(f) = n I_{n-1} \left(f(\cdot,t) \right),$$

where D is the derivative in the ℓ direction.

Remark 1.5 It is possible to compute the ℓ–th derivative as follows:

$$\begin{aligned}
D_t^{(\ell)} J_n^{(i_1,\ldots,i_n)}(f) &= \mathbf{1}_{\{i_n=\ell\}} J_{n-1}^{(i_1,\ldots,i_{n-1})} \left(f(\cdot,t) \mathbf{1}_{\Sigma_n^{(n)}(t)}(\cdot) \right) \\
&\quad + \int_t^\infty D_t^{(\ell)} J_{n-1}^{(i_1,\ldots,i_{n-1})} \left(f(\cdot,s) \mathbf{1}_{\Sigma_n^{(n)}(s)}(\cdot) \right) dH^{(i_n)}(s).
\end{aligned} \tag{1.3}$$

Note that (1.3) is a rough version of formula (1.46) of Nualart (1995) or Proposition 4.3 of Ma et al. (1998)

$$D_t\,(\delta(u)) = u_t + \int_0^\infty (D_t u_s)\, dW_s .$$

Now we define the spaces of the random variables that are differentiable in the ℓ-th direction. For this, we define the following subset of $L^2(\Omega)$:

$$\begin{aligned}\mathbb{D}^{(\ell)} \quad = \quad & \Bigg\{ F \in L^2(\Omega),\ F = EF + \sum_{n=1}^{\infty} \sum_{i_1,\ldots,i_n \geq 1} J_n^{(i_1,\ldots,i_n)}(f_{i_1,\ldots,i_n}) : \\ & \sum_{n=1}^{\infty} \sum_{i_1,\ldots,i_n \geq 1} \sum_{k=1}^{n} \mathbf{1}_{\{i_k=\ell\}} q_{i_1} \cdots \widehat{q_{i_k}} \cdots q_{i_n} \\ & \cdot \int_0^\infty \|f_{i_1,\ldots,i_n}(\cdots, t, \cdots)\mathbf{1}_{\Sigma_n^{(k)}(t)}\|^2_{L^2([0,\infty)^{n-1})}\, dt < \infty \Bigg\}\end{aligned}$$

Remark 1.6 Note that when the family $\{H^{(i)}, i \geq 1\}$ has only one element, then the expression for F becomes

$$F = EF + \sum_{n=1}^{\infty} J_n(f_n),$$

where, without loss of generality, we can assume the symmetry of $f_n \in L^2(\mathbb{R}_+^n)$, and therefore,

$$F = EF + \sum_{n=1}^{\infty} J_n(f_n) = EF + \sum_{n=1}^{\infty} I_n\left(\frac{1}{n!} f_n\right).$$

Further, from the symmetry of f_n, the expression in the definition of $\mathbb{D}^{(1)}$ is

$$\begin{aligned}\sum_{n=1}^{\infty}\sum_{k=1}^{n} q_1^{n-1} \int_0^\infty \int_{\Sigma_n^{(k)}(t)} \quad & f(\cdots, t, \cdots)^2 \mathbf{1}_{\Sigma_n^{(k)}}(t))\, dt \\ & = \sum_{n=1}^{\infty} n q_1^{n-1} \int_{\Sigma_n} f(t_1, \ldots, t_n)^2\, dt_1 \cdots dt_n \\ & = \sum_{n=1}^{\infty} n q_1^{n-1} \frac{1}{n!} \|f_n\|^2 = \sum_{n=1}^{\infty} n q_1^{n-1}\, n! \left\| \frac{1}{n!} f_n \right\|^2,\end{aligned}$$

which is the definition of $\mathbb{D}^{2,1}$ in the classical Malliavin calculus for Gaussian processes.

Definition 1.7 *Given* $F \in \mathbb{D}^{(\ell)}$ *such that*

$$F = EF + \sum_{n=1}^{\infty} \sum_{i_1,\ldots,i_n \geq 1} J_n^{(i_1,\ldots,i_n)}(f_{i_1,\ldots,i_n}),$$

we define the derivative of F in the ℓ-th direction as the element of $L^2(\Omega \times \mathbb{R}_+)$ given by

$$D_t^{(\ell)} F = \sum_{n=1}^{\infty} \sum_{i_1,\dots,i_n} \sum_{k=1}^{n} \mathbf{1}_{\{i_k=\ell\}} J_{n-1}^{(i_1,\dots,\widehat{i_k},\dots,i_n)} \left(f_{i_1,\dots,i_n}(\cdots, t, \cdots) \mathbf{1}_{\Sigma_n^{(k)}(t)}(\cdot) \right)$$

Observe that, as in the classical situation for Gaussian processes, $\mathbb{D}^{(\ell)}$ is dense in $L^2(\Omega)$, since the elements of $L^2(\Omega)$ with a finite chaotic expansion are in $\mathbb{D}^{(\ell)}$. Then, also as in the Brownian case, we can define the adjoint operator of $D^{(\ell)}$, $\delta^{(\ell)}$ which will be called the Skorohod integral in the ℓ–th direction. We will not continue this development here because we will only deal with the derivative operators.

From the chaotic representation property (Theorem 1.2) we can prove the following formula:

Theorem 1.8 (Clark-Ocone Formula) *Let $F \in \bigcap_{n=1}^{\infty} \mathbb{D}^{(n)}$. Then*

$$F = E[F] + \sum_{k=1}^{\infty} \int_0^{\infty} {}^p(D_t^{(k)} F)\, dH_t^{(k)},$$

where ${}^pG_t$ denotes the predictable projection of the process $\{G_t,\ t \geq 0\}$.

Proof By the chaotic decomposition property (Theorem 1.2), we have that F has the $L^2(\Omega)$ representation of the form

$$\begin{aligned} F &= E[F] + \sum_{k=1}^{\infty} \int_0^{\infty} \Bigg[f_k(t) \\ &\quad + \sum_{n=2}^{\infty} \sum_{i_1,\dots,i_{n-1} \geq 1} \int_0^{t^-} \int_0^{t_{n-1}^-} \cdots \int_0^{t_2^-} f_{i_1,\dots,i_{n-1};k}(t_1,\dots,t_{n-1},t) \\ &\quad \cdot dH^{(i_1)}(t_1) \cdots dH^{(i_{n-1})}(t_{n-1}) \Bigg] dH^{(k)}(t). \end{aligned}$$

The expression between brackets is

$${}^p(D_t^{(k)} F) = E\left[D_t^{(k)} F / \mathscr{F}_{t-} \right].$$

Note that only one summand of the derivative remains because the other ones cancel by the conditional expectation with respect to $\mathscr{F}_{t-}$. □

1.4 Lévy processes with a finite number of jump sizes

In the next section we consider a Lévy process for which the family $\{H^{(i)},\ i \geq 1\}$ has only a finite number of elements different than zero. We summarize here some results about this case. We recall that the family $\{H^{(i)},\ i \geq 1\}$ was introduced in (1.2).

Proposition 1.9 *Suppose that, for some $j \geq 1$, $H^{(j)} = 0$. Then the number of different jump sizes of X that are not zero is at most $j-1$.*

Proof First, we note that when $H^{(1)} = 0$, then $X_t = c\,t$, and the Proposition is true.

Now, assume that for some $j \geq 2$ we have $H_t^{(j)} = 0$. Then

$$Y_t^{(j)} = -\sum_{i=1}^{j-1} a_{j,i} Y_t^{(i)}.$$

It follows that

$$\Delta Y_t^{(j)} = \Delta X_t^{(j)} = -\sum_{i=1}^{j-1} a_{j,i} \Delta Y_t^{(i)} = -\sum_{i=1}^{j-1} a_{j,i} \Delta X_t^{(i)}.$$

Since $\Delta X_t^{(i)} = (\Delta X_t)^i$, we have that

$$(\Delta X_t)^j = -\sum_{i=1}^{j-1} a_{j,i} (\Delta X_t)^i .$$

This means that the size of the jump at t verifies the equation

$$z^j + \sum_{i=1}^{j-1} a_{j,i} z^i = 0,$$

which has no independent term, and then it has at most $j-1$ non zero different roots. □

The following is the reciprocal of the last proposition.

Proposition 1.10 *If the process X has only j different jump sizes, then*

1. *$H^{(k)} = 0, \forall k \geq j+1$, if X has no continuous part.*
2. *$H^{(k)} = 0, \forall k \geq j+2$, if X has continuous part.*

Proof Begin with the case where X has no continuous part. Let $c_1, \ldots, c_j$ be the different sizes of jumps. Observe that the Lévy-Itô representation of X is

$$X_t = \sum_{i=1}^{j} c_i N_t^{(i)} + \alpha t,$$

where $N^{(1)}, \ldots, N^{(j)}$ are independent Poisson processes and $\alpha \in \mathbb{R}$. It follows that

$$X_t^{(1)} = \sum_{i=1}^{j} c_i N_t^{(i)} + \alpha t \quad \text{and} \quad X_t^{(k)} = \sum_{i=1}^{j} c_i^k N_t^{(i)}, \; k \geq 2.$$

The vectors $(c_1, \dots, c_j), \dots, (c_1^j, \dots, c_j^j)$ form a base of $\mathbb{R}^j$ (their determinant is a Vandermonde determinant, which is non zero). As a consequence, for $k \geq j+1$, there exist $b_1, \dots, b_j \in \mathbb{R}$ (depending on k) such that

$$\left(c_1^k, \dots, c_j^k\right) = \sum_{i=1}^{j} b_i \, (c_1^i, \dots, c_j^i),$$

and

$$X_t^{(k)} = b_1(X_t^{(1)} - \alpha t) + \sum_{i=2}^{j} b_i \, X_t^{(i)}, \quad k \geq j+1.$$

Hence,

$$Y_t^{(k)} = \sum_{i=1}^{j} b_i \, Y_t^{(i)}, \quad k \geq j+1.$$

This implies that for $k \geq j+1$, $H^{(k)}$ is a linear combination of $Y^{(1)}, \dots, Y^{(j)}$. On the other hand, from the definition of $H^{(j)}$ it is easy to see (by induction and because of the fact that the coefficient of $Y^{(j)}$ in the expression of $H^{(j)}$ is 1) that we can express $Y^{(j)}$ as a linear combination of $H^{(1)}, \dots, H^{(j)}$. Then, we have that $H^{(k)}$ $(k \geq j+1)$ is a linear combination of $H^{(1)}, \dots, H^{(j)}$. From the orthogonality properties of the $H^{(i)}$ it follows that $H^{(k)} = 0$.

Consider now the case where X has a continuous part. In this case we know that

$$X - EX = X^c + X^d,$$

where X^c is a continuous martingale and X^d is a martingale given by

$$X_t^d = \sum_{i=1}^{j} c_i N_t^{(i)},$$

where $N^{(1)}, \dots, N^{(j)}$ are independent Poisson processes. Then,

$$X_t^{(k)} = \sum_{i=1}^{j} c_i^k N_t^{(i)}, \quad k \geq 2.$$

As before, we can write $X^{(k)}$ $(k \geq j+1)$ as

$$X_t^{(k)} = b_1 \left(X_t^d + E\left(\sum_{i=1}^{j} c_i N_t^{(i)} \right) \right) + \sum_{i=2}^{j} b_i X_t^{(i)}, \quad k \geq j+1.$$

Then,

$$\Delta X_t^{(k)} = b_1 \Delta X_t^d + \sum_{i=2}^{j} b_i \, \Delta X_t^{(i)}, \quad k \geq j+1.$$

Equivalently,

$$(\Delta X_t)^k = b_1\,\Delta X_t + \sum_{i=2}^{j} b_i\,(\Delta X_t)^i\,, \quad k \geq j+1,$$

and multiplying by ΔX_t, adding and centering, we obtain

$$Y_t^{(k+1)} = \sum_{i=1}^{j} b_i Y_t^{(i+1)}, \quad k \geq j+1,$$

and we deduce as in the first part of this proof that $H^{(k)} = 0,\ k \geq j+2$. □

Remark 1.11 A normal martingale X (see Ma et al. 1998) is a martingale such that $\langle X, X\rangle_t = t$ and that possesses the chaotic representation property (see also Dellacherie et al. 1992, p. 199). As a consequence of the last two propositions we obtain that only the Brownian motion and the (compensated) Poisson processes are normal martingales and Lévy processes.

Corollary 1.12 *Let X be a Lévy process that satisfies condition (1.1) and is also a normal martingale. Then X is a Brownian motion or a process of the form $\alpha N_t - t/\alpha$, where N_t is Poisson process of intensity λ and $\alpha = \pm 1/\sqrt{\lambda}$.*

Proof Since $X_0 = 0$ and X is a martingale, $E[X_t] = 0$. So $H^{(1)} = X$. By the chaotic representation property,

$$\begin{aligned} H_t^{(i)} &= \sum_{n=1}^{\infty} \int_0^t \int_0^{s_{n-1}-} \cdots \int_0^{s_2-} f_n(s_1, \ldots, s_n) dX(s_1) \cdots dX(s_n) \\ &= \sum_{n=1}^{\infty} \int_0^t \int_0^{s_{n-1}-} \cdots \int_0^{s_2-} f_n(s_1, \ldots, s_n) dH^{(1)}(s_1) \cdots dH^{(1)}(s_n). \end{aligned}$$

On the other side,

$$H_t^{(i)} = \int_0^t dH^{(i)}(s).$$

From the uniqueness in the Schoutens-Nualart representation (Theorem 1.2) it follows that $H^{(i)} = 0,\ i \geq 2$. By Proposition 1.9, X has at most one jump size. According to the Lévy-Itô representation we have

$$X_t = \sigma W_t + \alpha(N_t - \lambda t),$$

where W is a Brownian motion, N a Poisson process of parameter $\lambda > 0$ and α is the jump size. Hence,

$$H_t^{(2)} = -\frac{\alpha^3 \lambda \sigma}{\alpha^2 \lambda + \sigma^2}\, W_t + \left(-\frac{\alpha^4 \lambda}{\alpha^2 \lambda + \sigma^2} + \alpha^2\right)(N_t - \lambda t).$$

If $\alpha = 0$, then $X = W$ since $\langle X, X\rangle_t = t$. When $\alpha \neq 0$, it is easy to check that if $\sigma \neq 0$ then $H^{(2)} \neq 0$, and the result follows. □

2 Simple Lévy processes

In this section we deal with the simple Lévy process given by

$$X_t = \sigma W_t + \alpha_1 N_1(t) + \ldots + \alpha_k N_k(t), \quad t \geq 0. \tag{2.1}$$

where $\{W_t, t \geq 0\}$ is a standard Brownian motion, $\{N_j(t), t \geq 0\}$, $j = 1, \ldots, k$, are independent Poisson processes (and independent of Brownian motion) of parameter $\lambda_1, \ldots, \lambda_k$, respectively, $\sigma > 0$ and $\alpha_1, \ldots, \alpha_k$ are different non-null numbers. The Lévy measure of X is $\nu = \sum_{j=1}^k \lambda_j \delta_{\alpha_j}$ and satisfies the condition (1.1) of Nualart and Schoutens (2000) for the validity of the chaotic representation property.

However, in this context, in addition to the family $\{H^{(1)}, \ldots, H^{(k+1)}\}$, we also have the set of martingales: $\{W_t, N_1(t) - \lambda_1 t, \ldots, N_k(t) - \lambda_k t\}$. It seems sensible to use the last family instead of the former. Indeed observe that

$$X_t^{(1)} = X_t = \sigma W_t + \sum_{j=1}^k \alpha_j N_j(t)$$

and

$$X_t^{(i)} = \sum_{0<s\leq t} (\Delta X_s)^i = \sum_{j=1}^k \alpha_j^i N_j(t), \quad i \geq 2,$$

because two Poisson processes defined in the same filtration are independent if and only if they do not jump at the same time (see Bertoin 1996, p. 5). Since

$$Y_t^{(i)} = X_t^{(i)} - m_i\, t, \quad i \geq 1,$$

we have

$$Y_t^{(1)} = \sigma W_t + \sum_{j=1}^k \alpha_j \left(N_j(t) - \lambda_j t\right)$$

and

$$Y_t^{(i)} = \sum_{j=1}^k \alpha_j^i \left(N_j(t) - \lambda_j t\right), \; i \geq 2.$$

Then, the martingales $Y^{(i)}$, $i \geq 1$, are linear combinations of $W_t, N_1(t) - \lambda_1 t, \ldots, N_k(t) - \lambda_k t$. Since the martingales $H^{(i)}$ are a linear combination of $Y^{(1)}, \ldots, Y^{(i)}$ it follows that $H^{(i)}$ are also a linear combination of $W_t, N_1(t) - \lambda_1 t, \ldots, N_k(t) - \lambda_k t$. Therefore, we have a chaotic representation property in terms of the iterated integrals with respect W_t, $N_1(t) - \lambda_1 t, \ldots, N_k(t) - \lambda_k t$. Remember also that by Proposition 1.10, $H^{(i)} = 0$, for $i \geq k+2$. Further, since we are assuming that $\alpha_1, \ldots, \alpha_k$ are different, there is one and only one way to express $W_t, N_t^{(1)} - \lambda_1 t, \ldots, N_t^{(k)} - \lambda_k t$ as a linear combination of $Y^{(1)}, \ldots, Y^{(k+1)}$, and then is deduced the unicity in the chaotic representation property in terms of the W_t, $N_t^{(1)} - \lambda_1 t, \ldots, N_t^{(k)} - \lambda_k t$.

To unify the notations, we will write

$$\begin{aligned} G_0(t) &= W_t, \\ G_j(t) &= N_j(t) - \lambda_j\, t,\ j = 1, \ldots, k. \end{aligned}$$

Also, we will denote by $L_n^{(i_1,\ldots,i_n)}(f)$ the iterated integral of f with respect to $G_{i_1}, \ldots, G_{i_n}$:

$$L_n^{(i_1,\ldots,i_n)}(f) = \\ = \int_0^\infty \left(\int_0^{t_n^-} \cdots \left(\int_0^{t_2^-} f(t_1, \ldots, t_n) dG_{i_1}(t_1) \right) \cdots dG_{i_{n-1}}(t_{n-1}) \right) dG_{i_n}(t_n).$$

Thus, we have a chaotic representation property in terms of the G_i:

Proposition 2.1 *Let $F \in L^2(\Omega, \mathscr{F}, P)$. Then F has a representation of the form*

$$F = E[F] + \sum_{n=1}^{\infty} \sum_{0 \le i_1,\ldots,i_n \le k} L^{(i_1,\ldots,i_n)}(f_{i_1,\ldots,i_n}),$$

where $f_{i_1,\ldots,i_j} \in L^2(\Sigma_j)$.

Obviously, a predictable representation property of the following form also holds:

Proposition 2.2 *Let $F \in L^2(\Omega, \mathscr{F}, P)$. Then F admits a representation of the form*

$$F = E[F] + \int_0^\infty \phi_0(t)\, dW_t + \sum_{j=1}^{k} \int_0^\infty \phi_j(t) d(N_j(t) - \lambda_j\, t),$$

where $\phi_0, \ldots, \phi_k$ are predictable processes such that $\int_0^\infty E[\phi_j^2(t)]\, dt < \infty$.

Further, we can define derivatives in the directions $W_t, N_1(t) - \lambda_1 t, \ldots, N_k(t) - \lambda_k t$ through the iterated integrals, mimicking the definition given in Sect. 1.3. We will denote by $D^{(0)}, \ldots, D^{(k)}$ the derivatives in the directions $W_t, N_1(t) - \lambda_1 t, \ldots, N_k(t) - \lambda_k t$ respectively. So we have

Theorem 2.3 *Let $F \in \bigcap_{j=0}^{k} \mathbb{D}^{(j)}$. Then*

$$F = E[F] + \int_0^\infty {}^p(D_t^{(0)} F)\, dW_t + \sum_{j=1}^{k} \int_0^\infty {}^p(D_t^{(j)} F)\, d(N_j(t) - \lambda_j\, t).$$

2.1 Interpretation of the derivatives for a simple Lévy process

In this section we still work with the simple model (2.1). We interpret the operators $D^{(\ell)}$ appearing in Theorem 2.3 and show some formulas that have interesting applications. The main result is that in certain cases it is possible to compute the derivatives in the directions W, N_1, etc. following the classical rules for each case. We point out that this is a nice idea due to Løkka (1999) that we think is only true for certain types of Lévy processes, such as the ones that we are considering.

To simplify the notation we consider the case $k = 1$, with $\sigma = \alpha = 1$:

$$X_t = W_t + N_t.$$

We will prove that $D^{(0)}F$ (respectively $D^{(1)}F$) can be interpreted as the usual Brownian Malliavin derivative (resp., the Poisson Malliavin derivative). The aim of this part is to prove Proposition 2.4 that, roughly speaking, says that for $F = f(Z, Z')$, where Z only depends on $\{W_t\}$ and Z' only depends on $\{N_t\}$, under some conditions on $f(x, y)$ and Z, we have that

$$D^{(0)}F = \frac{\partial f}{\partial x}(Z, Z')\, D^{(0)}Z,$$

and for $D^{(1)}F$ we get the corresponding usual formulas for the Poisson case.

The proof is done in the canonical spaces. Fix $T > 0$. Let $\Omega_W = \mathscr{C}_0([0, T])$ be the space of continuous functions defined on $[0, T]$ that are 0 at 0, with the topology of the uniform convergence, $\mathscr{F}_W$ the Borel σ–algebra and P_W the probability on $\mathscr{F}_W$ that makes the projections $w_t : \Omega_W \to \mathbb{R}$ a Brownian motion.

On the other hand, consider the canonical space of a Poisson process (see Neveu 1977): Let $\Omega_N = \bigcup_{n=0}^{\infty}[0, T]^n$, where $[0, T]^0 = \{a\}$, where a is a distinguished element; $\mathscr{F}_N$ be the collection of $G \subset \Omega_N$ such that $G \cap [0, T]^n \in \mathscr{B}([0, T]^n)$, $n \geq 1$, which is a σ–algebra, μ^n the Lebesgue measure on $[0, T]^n$, $n \geq 1$, and $\mu^0 = \delta_a$. Define on $\mathscr{F}_N$ the probability

$$P_N(A) = \sum_{n=0}^{\infty} e^{-\lambda T}\frac{1}{n!}\lambda^n \mu^n\left(A \cap [0, T]^n\right).$$

Then, the following process:

$$n_t(a) = 0, \quad n_t(s_1, \ldots, s_n) = \sum_{j=1}^{n} \mathbf{1}_{[0,t]}(s_j),$$

is a Poisson process of parameter λ.

Now, consider $\left(\Omega_W \times \Omega_N, \mathscr{F}_W \otimes \mathscr{F}_N, P_W \times P_N\right)$ and put

$$\widetilde{w}_t(\omega, \omega') = w_t(\omega) \quad \text{and} \quad \widetilde{n}_t(\omega, \omega') = n_t(\omega').$$

Then it is clear that

$$\widetilde{w}_{\cdot} + \widetilde{n}_{\cdot} \overset{\mathscr{L}}{=} W_{\cdot} + N_{\cdot}.$$

From now on, we will write $W_t(\omega)$ and $N_t(\omega')$ instead of $\widetilde{w}_t(\omega, \omega')$ and $\widetilde{n}_t(\omega, \omega')$ respectively.

Since in $\Omega_W \times \Omega_N$ there is a product measure, there is an isometry (as Hilbert spaces)

$$L^2(\Omega_W \times \Omega_N) \simeq L^2(\Omega_W; L^2(\Omega_N)),$$

where

$$L^2(\Omega_W; L^2(\Omega_N)) = \{F : \Omega_W \to L^2(\Omega_N) : \int_{\Omega_W} \|F(\omega)\|^2_{L^2(\Omega_N)} \, dP_W(\omega) < \infty\}.$$

In the space $L^2(\Omega_W; L^2(\Omega_N))$ (which is essentially a Gaussian space), a derivative operator D can be defined in the usual Malliavin calculus sense. This operator is closed from $L^2(\Omega_W; L^2(\Omega_N))$ into $L^2(\Omega_W \times [0,T]; L^2(\Omega_N))$. Hence, if $F \in DomD$, we have that

$$\begin{aligned} DF \in L^2\left(\Omega_W; L^2\left(\Omega_N; L^2([0,T])\right)\right) &\simeq L^2\left(\Omega_W; L^2\left(\Omega_N \times [0,T]\right)\right) \\ &\simeq L^2\left(\Omega_W \times \Omega_N \times [0,T]\right). \end{aligned}$$

Now we are going to see that, through the identification of these spaces, the derivative $D^{(0)}$ in the direction W in the space $L^2(\Omega_W \times \Omega_N)$ defined in Sect. 1.3 is equivalent to the derivative operator D defined in $L^2(\Omega_W; L^2(\Omega_N))$. To prove this, it is sufficient to check the property for a dense subset of $\mathbb{D}^W$. The subspace of $\mathbb{D}^{(0)}$ that we consider is the set of linear combinations of iterated integrals of functions of $L^2([0,T]^n)$ of the form

$$f(t_1, \ldots, t_n) = f_1(t_1) \cdots f_n(t_n).$$

We will use the induction method on n. For $n = 1$, there are two cases:

(a) $F = \int_0^T f(r) \, dW_r$. Then,

$$D_t^{(0)} F = f(t),$$

and this is also the derivative of F as element of $L^2(\Omega_W; L^2(\Omega_N))$,

$$D_t F = f(t).$$

(b) $F = \int_0^T f(r) \, dN_r$. Then,

$$D_t^{(0)} F = 0,$$

On the other hand, F as a element of $L^2(\Omega_W; L^2(\Omega_N))$ is independent of ω and then we also have

$$DF = 0.$$

Assume that both operators agree for an iterated integral of order n. For an iterated integral of order $n+1$ we can also consider two cases.

(a) The first case is when F is a random variable of the form

$$\int_0^T \int_0^{t_{n+1}-} \cdots \int_0^{t_2-} f_1(t_1) \cdots f_{n+1}(t_{n+1}) \, dM^{(1)}(t_1) \cdots dM^{(n)}(t_n) dW(t_{n+1}),$$

where $M^{(1)}, \ldots, M^{(n)}$ can be either W or $N_t - \lambda t$. In this case, we can write F as

$$F = \int_0^T f_{n+1}(r)g(r-)\,dW_r,$$

where

$$g(r) = \int_0^r \int_0^{t_n-} \cdots \int_0^{t_2-} f_1(t_1)\cdots f_n(t_n)\,dM^{(1)}(t_1)\cdots dM^{(n)}(t_n).$$

Then, in agreement with Formula (1.3),

$$D_t^{(0)}F = f_{n+1}(t)g(t-) + \int_t^T f_{n+1}(r)D_t^{(0)}g(r-)\,dW_r.$$

On the other hand, it can be seen that $F = \int_0^T f_{n+1}(r)g(r-)\,dW_r$ is differentiable and that its derivative as a Skorohod integral (Nualart 1995, Formula (1.46) applied for ω' fixed) gives

$$\begin{aligned} D_t F &= f_{n+1}(t)g(t-) + \int_t^T D_t\left(f_{n+1}(r)g(r-)\right)\,dW_r \\ &= f_{n+1}(t)g(t-) + \int_t^T f_{n+1}(r)D_t g(r-)\,dW_r, \end{aligned}$$

and the equality between $D^{(0)}F$ and DF follows by the hypothesis of induction.

(b) Now we analyze the case where $M_t^{(n+1)} = \widehat{N}_t \underset{\text{def}}{=} N_t - \lambda t$. That is

$$\begin{aligned} F = & \int_0^T \int_0^{t_{n+1}-} \cdots \int_0^{t_2-} f_1(t_1)\cdots f_{n+1}(t_{n+1})\,dM^{(1)}(t_1)\cdots \\ & \cdot dM^{(n)}(t_n)d\widehat{N}(t_{n+1}), \end{aligned}$$

where, as in the last case, $M^{(1)},\ldots,M^{(n)}$ can be either W or $\widehat{N}_t = N_t - \lambda t$. Write

$$F = \int_0^T f_{n+1}(r)g(r-)\,d\widehat{N}_r,$$

where

$$\begin{aligned} g(r) &= \int_0^r \int_0^{t_n-} \cdots \int_0^{t_2-} f_1(t_1)\cdots f_n(t_n)\,dM^{(1)}(t_1)\cdots dM^{(n)}(t_n) \\ &= \int_0^r f_n(s)h(s-)\,dM^{(n)}(s), \end{aligned}$$

with

$$h(s) = \int_0^s \int_0^{t_{n-1}-} \cdots \int_0^{t_2-} f_1(t_1)\cdots f_{n-1}(t_{n-1})\,dM^{(1)}(t_1)\cdots dM^{(n-1)}(t_{n-1}).$$

By Formula (1.3), the derivative in the direction W is,

$$D_t^{(0)}F = \int_t^T f_{n+1}(r)\,D_t^{(0)}g(r-)\,d\widehat{N}(r). \tag{2.2}$$

To compute the derivative DF, first we use the integration by parts formula to obtain

$$F = \int_0^T f_{n+1}(r) g(r-)\, d\widehat{N}_r \tag{2.3}$$

$$\begin{aligned}
&= g(T-) \int_0^T f_{n+1}(r)\, d\widehat{N}_r - \int_0^T \left(\int_0^{r-} f_{n+1}(s)\, d\widehat{N}_s \right) dg(r) \\
&\quad - \left[\int_0^{\cdot} f_{n+1}(r)\, d\widehat{N}_r,\ g(\cdot) \right]_T \\
&= g(T-) \int_0^T f_{n+1}(r)\, d\widehat{N}_r - \int_0^T \left(f_n(r) h(r-) \int_0^{r-} f_{n+1}(s)\, d\widehat{N}_s \right) dM^{(n)}(r) \\
&\quad - \int_0^T f_{n+1}(r) f_n(r) h(r-)\, d\left[N, M^{(n)} \right]_r .
\end{aligned} \tag{2.4}$$

Now we divide this case in two steps:

(b1) If $M^{(n)} = W$, then the quadratic covariation in (2.4) is equal to 0. Hence,

$$\begin{aligned}
D_t F &= D_t \left(g(T-) \int_0^T f_{n+1}(r)\, d\widehat{N}_r \right) \\
&\quad - f_n(t) h(t-) \int_0^{t-} f_{n+1}(s)\, d\widehat{N}_s \\
&\quad - \int_0^T D_t \left(f_n(r) h(r-) \int_0^{r-} f_{n+1}(s)\, d\widehat{N}_s \right) dW(r) \\
&= (D_t g(T-)) \cdot \int_0^T f_{n+1}(r)\, d\widehat{N}_r && (2.5) \\
&\quad - f_n(t) h(t-) \int_0^{t-} f_{n+1}(s)\, d\widehat{N}_s && (2.6) \\
&\quad - \int_0^T \left(f_n(r) \int_0^{r-} f_{n+1}(s)\, dN_s \right) D_t h(r-)\, dW(r). && (2.7)
\end{aligned}$$

By hypothesis of induction, we can write in (2.5) and (2.7) $D^{(0)}$ instead of D. The operator $D^{(0)}$ has the following expected property:

$$\begin{aligned}
&D^{(0)} \left[\int_0^{t-} \int_0^{t_n-} \cdots \int_0^{t_2-} f_1(t_1) \cdots f_n(t_n)\, dM^{(1)}(t_1) \cdots dM^{(n)}(t_n) \right. \\
&\quad \left. \cdot \int_0^t f_{n+1}(s)\, dN(s) \right] \\
&= \int_0^t f_{n+1}(s)\, dN(s) && (2.8) \\
&\quad \cdot D^{(0)} \left[\int_0^{t-} \int_0^{t_n-} \cdots \int_0^{t_2-} f_1(t_1) \cdots f_n(t_n)\, dM^{(1)}(t_1) \cdots dM^{(n)}(t_n) \cdot \right], && (2.9)
\end{aligned}$$

that is, the factor that does not depend on ω behaves as a constant when we apply the operator $D^{(0)}$; formula (2.9) is proved developing the product by the integration by parts formula, depending on $M^{(n)} = W$ or $\widehat{N}$, and computing the derivative.

Finally, using (2.9) we can go from (2.5),(2.6) and (2.7) (with $D^{(0)}$ instead D) to (2.2) reversing the steps used to obtain (2.5),(2.6) and (2.7). This gives $DF = D^{(0)}F$.

(b2) When $M^{(n)} = \widehat{N}$, the proof is exactly as before, but here appears a new term because the last integral in (2.4) is

$$\int_0^T f_{n+1}(r) f_n(r) h(r-)\, d\left[\widehat{N}, M^{(n)}\right]_r = \int_0^T f_{n+1}(r) f_n(r-) h(r-)\, dN_r.$$

But this is an iterated integral of order n, and also we can apply the hypothesis of induction.

The same reasoning applies to the Poisson part of X_t, that is, the derivative $D^{(1)}F$ coincides with the *classical* Poisson derivative which can be defined in $L^2\left(\Omega_N; L^2(\Omega_W)\right)$.

The most important consequence of the proof that we have presented is that since in $L^2\left(\Omega_W; L^2(\Omega_N)\right)$ (respectively $L^2\left(\Omega_N; L^2(\Omega_W)\right)$) there are two equivalent definitions of the derivative operators (chaotic and directional), the same is true for the derivative $D^{(0)}$ (resp., for $D^{(1)}$, chaotic and difference operator). Therefore, in some cases we have very useful formulas. In the next proposition we summarize these results, based in Nualart (1995) and Nualart and Vives (1990).

Proposition 2.4 *Consider a simple Lévy process*

$$X_t = \sigma W_t + \alpha_1 N_1(t) + \ldots + \alpha_k N_k(t), \quad t \geq 0.$$

(a) Let $F = f(Z, Z') \in L^2(\Omega)$, where Z only depends on the Brownian motion W, and Z' only depends on the Poisson processes $N_1, \ldots, N_k$. Assume that $f(x, y)$ is a continuosly differentiable function with bounded partial derivatives in the variable x, and that $Z \in \mathbb{D}^{(0)}$. Then $F \in \mathbb{D}^{(0)}$ and

$$D^{(0)}F = \frac{\partial f}{\partial x}(Z, Z')\, D^{(0)}Z.$$

(b) Let $F \in \mathbb{D}^{(j)}$ for some $j \in \{1, \ldots, k\}$. Consider a generic element $\omega = (\omega_0, \ldots, \omega_k) \in \Omega = \Omega_W \times \Omega_{N_1} \times \cdots \times \Omega_{N_k}$, $\omega_j \in \bigcup_{n=0}^{\infty}[0, T]^n$. Then

$$D_t^{(j)}F(\omega) = F(\omega_0, \ldots, \omega_{j-1}, \omega_j + \delta_t, \omega_{j+1}, \ldots, \omega_k) - F(\omega),$$

where

$$\omega_j + \delta_t = \begin{cases} (s_1, \ldots, s_r, t), & \text{if } \omega_j = (s_1, \ldots, s_r), \\ t, & \text{if } \omega_j = \{a\}. \end{cases}$$

3 Approximation by simple Lévy processes

In this section we will consider a Lévy process of the form

$$X_t = \sigma W_t + \sum_{j=1}^{N_t} Z_j, \qquad t \geq 0, \tag{3.1}$$

(with the convention that the sum is 0 when $N_t = 0$) where

1. $\{W_t,\, t \geq 0\}$ is a standard Brownian motion.
2. $\{Z_n,\, n \geq 1\}$ is a sequence of i.i.d. square integrable random variables.
3. $\{W_t,\, t \geq 0\}$, $\{N_t,\, t \geq 0\}$ and $\{Z_n,\, n \geq 1\}$ are independent.

This process models the evolution of a particle following a Brownian motion and jumps at times given by a Poisson process of intensity λ, with jump sizes given by $Z_1, Z_2, \ldots$. The term $\{\sum_{j=1}^{N_t} Z_j,\, t \geq 0\}$ is a compound Poisson process. The generating triplet of $\{X_t,\ t \geq 0\}$ is $(\sigma, \lambda P_{Z_1}, 0)$ (see Sato 1999), where P_{Z_1} is the distribution of the jump size Z_1. The simple Lévy process (2.1) studied in Sect. 2 is a particular case of (3.1), when the random variable Z_1 is discrete and takes the values $\alpha_1, \ldots, \alpha_k$. We prove that, in the same way that every square integrable random variable can be approximated (in $L^2(\Omega)$) by simple random variables (taking only a finite number of different values), a Lévy process of type (3.1) can be approximated by simple Lévy processes in $L^2(\Omega \times [0, T])$.

Theorem 3.1 *Let $T > 0$. For each n, there exists a family of independent Poisson processes (and independent of $\{W_t,\, t \geq 0\}$), $\{N_1^n(t),\, t \geq 0\}, \ldots, \{N_{k_n}^n(t),\, t \geq 0\}$, and a family of constants $\alpha_1^n, \ldots, \alpha_{k_n}^n$ such that the Lévy process*

$$X^n(t) = \sigma W_t + \sum_{r=1}^{k_n} \alpha_r^n N_r^n(t), \qquad t \in [0, T], \tag{3.2}$$

converges to $\{X_t,\, t \geq [0, T]\}$ in $L^2(\Omega \times [0, T])$.

Proof It is enough to consider the jumps part of $X^{(n)}$ and X. The idea of the proof is to approximate the random variable Z_j by a sequence of simple random variables. Consider a standard approximation of Z_1 by simple functions; that is,

$$Z_1^{(n)} = \sum_{r=1}^{k_n} \alpha_r^n \mathbf{1}_{\{Z_1 \in A_r^n\}},$$

where $\mathscr{P}_n = \{A_1^n, \ldots, A_{k_n}^n\}$, $n \geq 1$, is a sequence of partitions of $\mathbb{R} - (-1/n, 1/n)$ and $\{\alpha_j^n,\, j = 1 \ldots, k_n\}$, $n \geq 1$ are numbers such that

$$|Z_1^{(n)}| \leq |Z_1|, \quad \forall n \geq 1,$$

and

$$\lim_n Z_1^{(n)} = Z_1 \quad \text{in } L^2(\Omega).$$

Now consider the approximation of every Z_j based in the same partition $\mathscr{P}_n$ and numbers α_r^n:

$$Z_j^{(n)} = \sum_{r=1}^{k_n} \alpha_r^n \mathbf{1}_{\{Z_j \in A_r^n\}}.$$

Note that

$$\sum_{j=1}^{N_t} Z_j^{(n)} = \sum_{j=1}^{N_t} \sum_{r=1}^{k_n} \alpha_r^n \mathbf{1}_{\{Z_j \in A_r^n\}} = \sum_{r=1}^{k_n} \alpha_r^n \sum_{j=1}^{N_t} \mathbf{1}_{\{Z_j \in A_r^n\}}.$$

Write

$$N_r^n(t) = \sum_{j=1}^{N_t} \mathbf{1}_{\{Z_j \in A_r^n\}}, \ \ r = 1, \dots, k_n,$$

and observe that they are Poisson processes of parameter $\lambda_r^n = \lambda P\{Z_1 \in A_r^n\}$. Further, note that they are independent since two of them do not jump at the same time (see Bertoin 1996, p. 5). Indeed, consider the first jump of N_t: only one of $N_1^n, \dots, N_{k_n}^n$ will jump depending on if Z_1 is in $A_1^n, \dots A_{k_n}^n$ etc.; the second jump will produce a jump on some of $N_1^n, \dots, N_{k_n}^n$, depending on Z_2. In general, every jump of N_t will produce only one jump of one of $N_1^n, \dots, N_{k_n}^n$.

In order to prove the theorem we need only to show that

$$\lim_n \sum_{r=1}^{k_n} \alpha_r^n N_r^n(t) = \sum_{j=1}^{N_t} Z_j, \quad \text{in } L^2(\Omega \times [0, T]).$$

This is done in the following way:

$$\begin{aligned}
\int_0^T E\left[\left(\sum_{r=1}^{k_n} \alpha_r^n N_r^n(t) - \sum_{j=1}^{N_t} Z_j\right)^2\right] dt &= \int_0^T E\left[\left(\sum_{j=1}^{N_t} \left(Z_j^{(n)} - Z_j\right)\right)^2\right] dt \\
&= \int_0^T \sum_{m=1}^{\infty} E\left[\left(\sum_{j=1}^{m} \left(Z_j^{(n)} - Z_j\right)\right)^2\right] e^{-\lambda t}\, \frac{\lambda^m t^m}{m!}\, dt \\
&\overset{(1)}{\leq} \int_0^T \left(\sum_{m=1}^{\infty} m \sum_{j=1}^{m} E\left[\left(Z_j^{(n)} - Z_j\right)^2\right]\right) e^{-\lambda t}\, \frac{\lambda^m t^m}{m!}\, dt \\
&= \int_0^T E\left[\left(Z_1^{(n)} - Z_1\right)^2\right] \left(\sum_{m=1}^{\infty} m^2\, e^{-\lambda t}\, \frac{\lambda^m t^m}{m!}\right) dt \\
&= E\left[\left(Z_1^{(n)} - Z_1\right)^2\right] \int_0^T (1 + \lambda t)\lambda t\, dt \underset{n\to\infty}{\longrightarrow} 0
\end{aligned}$$

where in step (1) we used the inequality $\left(\sum_{i=1}^m a_i\right)^2 \leq m \sum_{i=1}^m a_i^2$. $\square$

Remark 3.2 In the next section we will work with the Lévy process

$$X_t = (\mu - \lambda\kappa)t + \sigma W_t + \sum_{j=1}^{N_t} Z_j, \qquad t \geq 0, \tag{3.3}$$

where $\{W_t,\, t \geq 0\}$, $\{N_t,\, t \geq 0\}$ and $Z_1, Z_2, \ldots$ are as before, $\kappa = E[Z_1]$ and μ is a drift of the Brownian part. With the same notations as in the proof of Theorem 3.1 we have

$$E[Z_1^{(n)}] \underset{n\to\infty}{\longrightarrow} E[Z_1] = \kappa,$$

which implies

$$\lim_n \sum_{r=1}^{k_n} \alpha_r^n \lambda_r^n = \kappa\lambda.$$

Then, it follows that the approximation

$$X_t^{(n)} = \mu t + \sigma W_t + \sum_{r=1}^{k_n} \alpha_r^n \left(N_r^n(t) - \lambda_r^n t\right), \qquad t \in [0, T], \tag{3.4}$$

converges to the process $\{X_t,\, t \in [0, T]\}$ given in (3.3) in $L^2\left(\Omega \times [0, T]\right)$.

4 Option hedging in a market with jumps

Here we consider a market with only one riskless asset, $A(t)$, determined by

$$\begin{aligned} dA(t) &= r\,A(t)\,dt \\ A(0) &= 1 \end{aligned}$$

where $r > 0$ is a constant. Also consider an asset with risk, $S(t)$, satisfying the equation

$$\begin{aligned} dS(t) &= S(t-)\,dX(t) \\ S(0) &= s_0 \end{aligned}$$

(s_0 is a constant) where $X(t)$ is the Lévy process defined in (3.3) with the condition that $Z_n > -1$, $n \geq 1$, which implies that the asset prices will be always non negative.

4.1 Approximative market

Our objective is to hedge an european call based on the asset S with maturity T and strike price $K > 0$; the final profit is

$$U = (S(T) - K)^{+} .$$

Since U is a square integrable random variable, by Theorem 1.3 it can be represented as a (possibly) infinite sum of stochastic integrals with respect to the family $\{H^{(i)}, i \geq 1\}$ related to X. In order to carry out a realistic hedging, we need to do two things: to consider a finite sum and to be able to compute the integrands. Both things can be done approximating the Lévy process X by a simple process, which is possible thanks to Theorem 3.1 and Remark 3.2. Therefore, we will consider the approximating process

$$X^n(t) = \mu t + \sigma W_t + \sum_{s=1}^{k_n} \alpha_s^n \left(N_s^n(t) - \lambda_s^n t \right) ,$$

where $\alpha_s^n > -1, \; s = 1, \ldots, k_n$. The market modeled by this process will be called approximative market.

Lemma 4.1 *Let $\{S^n(t), t \in [0, T]\}$, be the solution of the equation*

$$\begin{aligned} dS^n(t) &= S^n(t-)\, dX^n(t) \\ S^n(0) &= s_0 \end{aligned}$$

Then

$$\lim_n S^n = S, \quad \textit{in } L^2(\Omega \times [0, T]).$$

Proof Note that $S(t)$ and $S^n(t)$ are given by

$$\begin{aligned} S(t) &= s_0 \exp\{Ct + \sigma W_t\} \prod_{j=1}^{N_t} (1 + Z_j), \\ S^n(t) &= s_0 \exp\{C_n t + \sigma W_t\} \prod_{j=1}^{N_t} (1 + Z_j^{(n)}), \end{aligned}$$

where $\lim_n C_n = C$. So

$$\begin{aligned} &\int_0^T E\left[(S^n(t) - S(t))^2 \right] dt \\ &\leq 2s_0^2 \int_0^T E\left[\left\{ |\exp\{C_n t\} - \exp\{Ct\}| \exp\{\sigma W_t\} \prod_{j=1}^{N_t} (1 + Z_j^{(n)}) \right\}^2 \right] dt \quad (4.1) \end{aligned}$$

$$+2s_0^2 \int_0^T E\left[\left\{\exp\{Ct\}\exp\{\sigma W_t\}\left(\prod_{j=1}^{N_t}(1+Z_j^{(n)})\right.\right.\right.$$
$$\left.\left.\left.-\prod_{j=1}^{N_t}(1+Z_j)\right)\right\}^2\right] dt \tag{4.2}$$

Consider the term (4.1). We have that

$$E[\exp\{2\sigma W_t\}] = \exp\{2\sigma^2 t\} \leq C_1', \quad t \in [0,T],$$

where C_1' is a constant. Also,

$$\begin{aligned} E\left[\prod_{j=1}^{N_t}(1+Z_j^{(n)})^2\right] &= \sum_{m=0}^{\infty}\left(E\left[(1+Z_1^{(n)})^2\right]\right)^m e^{-\lambda t}\frac{(\lambda t)^m}{m!} \\ &= \exp\left\{\lambda t\left(E\left[\left(1+Z_1^{(n)}\right)^2\right]-1\right)\right\} \leq C_2', \quad t\in[0,T], \end{aligned}$$

due to the fact that $|Z_1^{(n)}| \leq |Z_1|$. Hence, using the independence of the factors, we obtain that (4.1) is bounded by

$$2s_0^2 C_1' C_2' \int_0^T |\exp\{C_n t\} - \exp\{Ct\}|\, dt$$

which goes to zero as $n \to \infty$.

The term (4.2) is bounded by

$$C_3' \int_0^T E\left[\left(\prod_{j=1}^{N_t}(1+Z_j^{(n)}) - \prod_{j=1}^{N_t}(1+Z_j)\right)^2\right] dt. \tag{4.3}$$

Developing the expectation as above, and from the formula

$$\prod_{i=1}^m a_i - \prod_{i=1}^m b_i = \sum_{i=1}^m a_1\cdots a_{i-1}(a_i - b_i)b_{i+1}\cdots b_m$$

and Cauchy inequality $\left(\sum_{i=1}^m a_i\right)^2 \leq m\sum_{i=1}^m a_i^2$, (4.3) is bounded by $C_4' E[(Z_1^{(n)} - Z_1)^2]$, that goes to zero. □

Proposition 4.2 *Let $U_n = (S^n(T) - K)^+$. Then U_n converges to U in $L^2(\Omega)$.*

Proof The proof follows the same steps of Lema 4.1. □

To simplify the notation, we will omit the index n in the expressions of $X^n(t)$, $S^n(t)$, U_n, α_r^n and λ_r^n. Then, the approximate price $S(t)$ will be driven by the simple Lévy process

$$X(t) = \mu t + \sigma W_t + \sum_{r=1}^{k} \alpha_r \left(N_r(t) - \lambda_r t\right).$$

We will also denote by S_t^* and U^* the discounted price:

$$S_t^* = e^{-rt} S_t \quad \text{and} \quad U^* = e^{-rt} U.$$

4.2 Option pricing in the approximative market

It is well-known that the problem of pricing an option consists in finding a unique equivalent measure Q that makes S_t^* a martingale. In this context this problem has been solved by Jensen (1999) assuming the existence of k additional assets, defined by equations

$$dP_j(t) = P_j(t-)\, dX_j(t), \quad j = 1, \ldots, k,$$

where

$$X_j(t) = \mu_j t + \sigma_j W_t + \sum_{r=1}^{k} \alpha_{j,r} \left(N_r(t) - \lambda_r t\right).$$

(We will return later to these μ_j, σ_j and $\alpha_{j,r}$).

This is a known idea: to hedge an option when there are $k + 1$ sources of randomness $(W, N_1, \ldots, N_k)$, we need $k + 1$ assets so that the market be complete. Using the Girsanov Theorem (see Sato 1999), Jensen (1999) proves that a condition for the existence of one and only one equivalent probability Q (that depens on k) such that the discounted prices $S^*, P_1^*, \ldots, P_k^*$ $(P_j^*(t) = e^{-rt} P_j(t))$ are Q–martingales is that there exist M, $L_1, \ldots, L_k$,with $L_j \leq \lambda_j$ such that

$$\begin{pmatrix} \sigma & \alpha_1 & \cdots & \alpha_k \\ \sigma_1 & \alpha_{11} & \cdots & \alpha_{1k} \\ \vdots & & & \vdots \\ \sigma_k & \alpha_{k1} & \cdots & \alpha_{kk} \end{pmatrix} \begin{pmatrix} M \\ L_1 \\ \vdots \\ L_k \end{pmatrix} = \begin{pmatrix} \mu - r \\ \mu_1 - r \\ \vdots \\ \mu_k - r \end{pmatrix}. \tag{4.4}$$

From our point of view we can present another explanation of these results. The Clark-Ocone formula given in Theorem 2.3 can be applied to the approximated marked. In particular, this implies that every square integrable random variable of the type $f(S_T)$ can be represented as a sum of $k+1$ stochastic integrals (respect to $W, N^{(1)}, \ldots, N^{(k)}$). However, what we need is a representation using the assets S, $P_1, \ldots, P_k$ as integrators; in other words, that the Q-martingales (discounted) S^*, $P_1^*, \ldots, P_k^*$ be a base for the representation. As we will see, the condition we get for that is exactly Jensen's condition.

We skip the details but we keep the result (see Jensen 1999) that under condition (4.4) there is one and only one equivalent probability Q, a Q–Brownian motion W^Q and k independent Q–Poisson processes of parameter $\widetilde{\lambda}_j = \lambda_j - L_j$, $N_1^Q, \ldots, N_k^Q$, independent of W^Q, such that we can write

$$\begin{aligned}
\frac{dS(t)}{S(t-)} &= r\,dt + \sigma dW^Q(t) + \sum_{r=1}^{k} \alpha_r \left(dN_r^Q(t) - \widetilde{\lambda}_r dt\right) \\
\frac{dP_1(t)}{P_1(t-)} &= r\,dt + \sigma_1 dW^Q(t) + \sum_{r=1}^{k} \alpha_{1,r} \left(dN_r^Q(t) - \widetilde{\lambda}_r dt\right) \\
&\vdots \\
\frac{dP_k(t)}{P_k(t-)} &= r\,dt + \sigma_k dW^Q(t) + \sum_{r=1}^{k} \alpha_{k,r} \left(dN_r^Q(t) - \widetilde{\lambda}_r dt\right)
\end{aligned} \tag{4.5}$$

The price of an european call $U = (S(T) - K)^+$ obtained by Jensen (1999) is

$$\begin{aligned}
c_0 &= \exp\left\{T \sum_{j=1}^{k} \widetilde{\lambda}_j\right\} \sum_{n_1,\ldots,n_k=1}^{\infty} \prod_{j=1}^{k} \frac{\left[(1+\alpha_j)\widetilde{\lambda}_j T\right]^{n_j}}{n_j!} \\
&\quad \cdot \left(S_0 \exp\left\{-T \sum_{j=1}^{k} \alpha_j \widetilde{\lambda}_j\right\} N\left(d_1(n_1,\ldots,n_k;0;S_0)\right)\right. \\
&\quad \left. - \frac{e^{-rT}K}{\prod_{j=1}^{k}(1+\alpha_j)^{n_j}} N\left(d_2(n_1,\ldots,n_k;0;S_0)\right)\right)
\end{aligned} \tag{4.6}$$

where $N(z)$ denotes the distribution function at point z of a standard normal random variable,

$$\begin{aligned}
d_1(n_1,\ldots,n_k;t;x) &= \frac{1}{\sigma\sqrt{T-t}} \left[\ln\frac{x}{K} + \left(r + \frac{1}{2}\sigma^2 - \sum_{j=1}^{k} \alpha_j \widetilde{\lambda}_j\right)(T-t)\right] \\
&\quad + \frac{1}{\sigma\sqrt{T-t}} \sum_{j=1}^{k} n_j \ln(1+\alpha_j)
\end{aligned} \tag{4.7}$$

and

$$d_2(n_1,\ldots,n_k;t;x) = d_1(n_1,\ldots,n_k;t;x) - \sigma\sqrt{T-t}. \tag{4.8}$$

(Remember that $x > 0$ and $\alpha_j > -1,\ j = 1,\ldots,k$).

For practical reasons, the infinite sum in Formula (4.6) needs to be truncated and Jensen (1999) carried out a simulation study to analyze the convenient level of truncation and the degree of the approximation.

4.3 Option hedging in the approximative marked

By Clark-Ocone Formula (Theorem 2.3) we have

$$U^* = E_Q[U^*] + \int_0^T \phi_0(t)\, dW^Q(t) + \sum_{j=1}^{k} \int_0^T \phi_j(t)\, d(N_j^Q(t)(t) - \widetilde{\lambda}_j t).$$

where

$$\phi_j(t) = E_Q\left[D_t^{(j)} U^* / \mathscr{F}_{t-}\right], \quad j = 0, \dots, k,$$

that we will calculate below.

The explicit solution of the first equation in (4.6) is

$$S_t = S_0 exp\{Ct + \sigma W_t^Q\} \prod_{r=1}^{k} (1 + \alpha_r)^{N_r(t)},$$

where $C = r - \frac{1}{2}\sigma^2 - \sum_{r=1}^{k} \alpha_r \widetilde{\lambda}_r$. To compute $D_t^{(0)}\left[(S(T) - K)^+\right]$ we use the standard procedure of approximating the function $(x - K)^+$ by a sequence of C^1 functions (see Øksendal 1996 for details), and Proposition 2.4 (a); thus we obtain that

$$D_t^{(0)}\left[(S(T) - K)^+\right] = \sigma S(T) \mathbf{1}_{[K,\infty)}(S(T)).$$

Now we want to compute $E[f(S_T)/\mathscr{F}_{t-}]$, where $f(y) = y\mathbf{1}_{[K,\infty)}(y)$ and $K > 0$. For this, we first calculate $E[f(S_T)/\mathscr{F}_s]$. Now, we decompose S_T as follows:

$$S_T = S_s R_s,$$

where

$$R_s = \exp\{C(T - s) + \sigma(W_T^Q - W_s^Q)\} \prod_{j=1}^{k} (1 + \alpha_j)^{N_j(T) - N_j(s)}.$$

Since R_s is independent of S_s and $\mathscr{F}_s$,

$$E[f(S_T)/\mathscr{F}_s] = E[f(S_s R_s)/\mathscr{F}_s] = E[f(x R_s)]|_{x=S_s}.$$

Note that R_s is a function of a Gaussian random variable and k Poisson random variables, which are independent. Hence, computing first the expectation with respect to the Gaussian random variable and after with respect to the Poisson random variables, we get the following explicit expression:

$$\begin{aligned} E[f(x R_s)] &= x \exp\left\{\left(r + \sum_{j=1}^{k} (\alpha_j - 1)\widetilde{\lambda}_j\right)(T - s)\right\} \\ &\quad \cdot \sum_{n_1,\dots,n_k=1}^{\infty} \prod_{j=1}^{k} \frac{\left[(1 + \alpha_j)(\widetilde{\lambda}_j(T - s))\right]^{n_j}}{n_j!} N\left(d_1(n_1, \dots, n_k; s; x)\right), \end{aligned}$$

where $d_1(n_1, \dots, n_k; s; x)$ is given in (4.7).

Finally, since $\mathscr{F}_{t-} = \bigvee_{s<t} \mathscr{F}_s$ we can apply a theorem of convergence of martingales (see Dellacheie and Meyer 1980, p. 28)

$$E[f(S_T)/\mathscr{F}_{t-}] = \lim_{s_n \uparrow t} E[f(S_T)/\mathscr{F}_{s_n}], \quad \text{a.s.}$$

Given that the function $N\,(d_1(n_1,\ldots,n_k;s;x))$ is continuous at $(s,x) \in (0,T) \times (0,\infty)$, by the Weierstrass M–test and the fact that $\{S_t,\, t \geq 0\}$ is cadlag, we obtain

$$\begin{aligned}\phi_0(t) &= \sigma E[f(S_T)/\mathscr{F}_{t-}] = \sigma S_{t-} \exp\left\{\left(r + \sum_{j=1}^{k}(\alpha_j - 1)\widetilde{\lambda}_j\right)(T-t)\right\} \\ &\quad \cdot \sum_{n_1,\ldots,n_k=1}^{\infty} \prod_{j=1}^{k} \frac{\left[(1+\alpha_j)(\widetilde{\lambda}_j(T-t))\right]^{n_j}}{n_j\,!} N\left(d_1(n_1,\ldots,n_k;t;S_{t-})\right).\end{aligned}$$

The derivative $D_t^{(j)}\left[(S(T)-K)^+\right]$, $j = 1,\ldots,k$ is calculated using Proposition 2.4 (b) and we obtain

$$D_t^{(j)}\left[(S(T)-K)^+\right] = \left((\alpha_j+1)\,S(T)-K\right)^+ - (S(T)-K)^+ .$$

The conditional expectation is computed following the same techniques as above. We obtain the expression

$$\begin{aligned}&\phi_j(t) = E\left[D_t^{(j)}(S(T)-K)^+/\mathscr{F}_{t-}\right] = \\ &(1+\alpha_j)S_{t-} \exp\left\{\left(r + \sum_{j=1}^{k}(\alpha_j - 1)\widetilde{\lambda}_j\right)(T-t)\right\} \\ &\cdot \sum_{n_1,\ldots,n_k=1}^{\infty} \prod_{j=1}^{k} \frac{\left[(1+\alpha_j)(\widetilde{\lambda}_j(T-t))\right]^{n_j}}{n_j\,!} N\left(d_1(n_1,\ldots,n_k;t;(1+\alpha_j)S_{t-})\right) \\ &-e^{-\sum_{j=1}^{k}\widetilde{\lambda}_j(T-t)} \sum_{n_1,\ldots,n_k=1}^{\infty} \prod_{j=1}^{k} \frac{\left[(\widetilde{\lambda}_j(T-t))\right]^{n_j}}{n_j\,!} \\ &\qquad \cdot N\left(d_2(n_1,\ldots,n_k;t;(1+\alpha_j)S_{t-})\right)\end{aligned}$$

$$\begin{aligned}&-S_{t-} \exp\left\{\left(r + \sum_{j=1}^{k}(\alpha_j - 1)\widetilde{\lambda}_j\right)(T-t)\right\} \\ &\quad \cdot \sum_{n_1,\ldots,n_k=1}^{\infty} \prod_{j=1}^{k} \frac{\left[(1+\alpha_j)(\widetilde{\lambda}_j(T-t))\right]^{n_j}}{n_j\,!} N\left(d_1(n_1,\ldots,n_k;t;S_{t-})\right) \\ &+e^{-\sum_{j=1}^{k}\widetilde{\lambda}_j(T-t)} \sum_{n_1,\ldots,n_k=1}^{\infty} \prod_{j=1}^{k} \frac{\left[(\widetilde{\lambda}_j(T-t))\right]^{n_j}}{n_j\,!} N\left(d_2(n_1,\ldots,n_k;t;S_{t-})\right),\end{aligned}$$

where $d_1(n_1,\dots,n_k;t;x)$ and $d_2(n_1,\dots,n_k;t;x)$ are given in (4.7) and (4.8) respectively.

The last step is to write U^* as a sum of integrals with respect the assets S^*, $P_1^*,\dots,P_k^*$. Denote by B the matrix

$$B = \begin{pmatrix} \sigma & \alpha_1 & \cdots & \alpha_k \\ \sigma_1 & \alpha_{11} & \cdots & \alpha_{1k} \\ \vdots & & & \vdots \\ \sigma_k & \alpha_{k1} & \cdots & \alpha_{kk} \end{pmatrix}$$

The system (4.5) is

$$\begin{pmatrix} \frac{dS^*(t)}{S^*(t-)} \\ \frac{dP_1^*(t)}{P_1^*(t-)} \\ \vdots \\ \frac{dP_k^*(t)}{P_k^*(t-)} \end{pmatrix} = B \begin{pmatrix} dW_t^Q \\ d(N_1(t)^Q - \widetilde{\lambda}_1 t) \\ \vdots \\ d(N_k(t)^Q - \widetilde{\lambda}_k t) \end{pmatrix}$$

It follows that if B is invertible (which is Jensen's condition) we can define

$$(\psi_0(t),\dots,\psi_k(t)) = (\phi_0(t),\dots,\phi_k(t))\, B^{-1} \begin{pmatrix} \frac{1}{S^*(t-)} & 0 & \cdots & 0 \\ 0 & \frac{1}{P_1^*(t-)} & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \frac{1}{P_k^*(t-)} \end{pmatrix}$$

and we obtain the hedging

$$U^* = E_Q[U^*] + \int_0^T \psi_0(t)\, dS^*(t) + \sum_{j=1}^k \int_0^T \psi_j(t)\, dP_j^*(t). \tag{4.9}$$

Note that the Black-Scholes model is a special case assuming that all the jump sizes are zero: $\alpha_1 = \cdots = \alpha_k = 0$. Then $\phi_1 = \cdots = \phi_k = 0$ and formula (4.9) becomes a Black-Scholes hedging.

Acknowledgements. We would like to thank Professor David Nualart for fruitful suggestions. J.A. León thanks the hospitality of the Universitat Autònoma de Barcelona during his visit. This work was supported by grants BFM2000-9 by DGES and 1999SGR-87 by CIRIT.

References

Aase, K, Øksendal, B., Privault, N., Ubøe, J.: White noise generalizations of the Clark-Haussmann-Ocone theorem with appications to mathematical finance. Finance Stochast. **4** (4), 465–496 (2000)

Bertoin, J.: Lévy Processes. Cambridge: Cambridge University Press 1996

Black, F., Scholes, M.: The pricing of options and corporate liabilities. Journal of Political Economy **81**, 637–653 (1973)

Dellacherie, C., Meyer, P. A.: Probabilités et potentiel. Theorie des martingales. Paris: Hermann 1982

Dellacherie, C., Maissonneuve, B. Meyer, P. A.: Probabilités et potentiel. Theorie de potentiel. Processus de Markov. Paris: Hermann 1992

Jensen, B.: Option pricing in the jump–diffusion model with a random jump amplitude: A complete market approach. Centre for Analytical Finance. Working Paper Series, 42 (1999)

Løkka, A.: Martingale representation, chaos expansion and Clark–Ocone formula. MaPhySto, Res. Report 22 (1999)

Ma, J., Protter, P., San Martin, J.: Anticipating integrals for a class of martingales. Bernoulli **4** (1), 81–114 (1998)

Merton, R.: Option pricing when underlying stock returns are discontinuous. J. Financial Econ. **3**, 125–144 (1976)

Nualart, D.: The Malliavin Calculus and related topics. Berlin Heidelberg New York: Springer 1995

Nualart, D., Schoutens, W.: Chaotic and predictable representation for Lévy processes. Stochast. Proc. Appl. **90** (1), 109–122 (2000)

Nualart, D., Vives, J.: Anticipative calculus for the Poisson Process based on the Fock space. In: Séminaire de Probabilités XXIV (Lecture Notes in Mathematics, 1426). Berlin Heidelberg New York: Springer 1990, pp. 154–165

Neveu, J.: Processus pontuels. In: École d'Eté de Probabilités de Saint Flour, VI (Lecture Notes in Mathematics, 598). Berlin Heidelberg New York: Springer 1977

Øksendal, B.: An introduction to Malliavin Calculus with applications to economics. Norges Handelshøyskole, Working Paper 3/96 (1996)

Sato, K-I.: Lévy processes and infinitely divisible distributions. Cambridge: Cambridge University Press 1999